

JAKE ROME

Jake-Rome.com

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Aerospace Engineering; Cumulative GPA: 4.00/4.00

Atlanta, GA

Aug. 2024 – Dec 2027

University of Minnesota

Dual Enrollment; Cumulative GPA: 4.00/4.00

Minneapolis, MN

Sept. 2023 – May. 2024

EXPERIENCE

UpperStory

Summer 2024 & Summer 2025

Research and Development Intern

St. Paul, MN

- Integrated a new game remote bumper system into a by iterating components, and updating PCB layouts and code
- Developed a compact vacuum tray for game components, reducing material usage through rapid prototyping
- Designed a custom battery and PCB while coordinating vehicle frame changes for electromechanical integration
- Engineered a 4WD wind-up car drivetrain, improving performance using design iteration and resin 3D printing

NASA L'SPACE NPWEE

Jan. 2025 – May. 2025

Project Manager and Primary Reviewer

Atlanta, GA

- Led a team in developing a proposal on spacecraft debris protection, managing project risk, budget, and timeline
- Learned to properly write and select an idea for a NASA proposal from NASA scientists, engineers, and managers
- Chaired a proposal review session, grading another team's proposal on the same criteria as official NASA proposals

EXTRACURRICULAR ACTIVITIES

Design Build Vertical Flight Club

Aug. 2025 – Current

Chief Engineer

Atlanta, GA

- Led technical direction as Chief Engineer, running weekly design reviews and coordinating unique 6 sub-teams
- Reduced drone airframe mass by 20% using SolidWorks topology optimization and lightweight structural redesign
- Designed a payload pickup mechanism using linear rails and computer vision to acquire and secure a 5-lb payload

Yellow Jacket Space Program

Jan. 2025 – Aug. 2025

Darcy Rocket Structures Team

Atlanta, GA

- Engineered and designed a mold for the Darcy nose cone, which was optimized for manufacturing constraints
- 3D printed the molds with ABS and used acetone surface treatment to create a smooth finish for the layup process

FIRST Tech Challenge Robotics

Sept. 2019 – Feb. 2024

Team Captain

Woodbury, MN

- Led the team to appear in 3 consecutive state championship tournaments, winning awards in each appearance
- Directed design and manufacturing sub-teams, introducing new systems to produce 160+ 3D printed parts
- Received the Dean's List Finalist award and represented Minnesota at the World Championship tournament

PROJECTS

- **3D Printed Clock:** Engineered and assembled a fully 3D printed clock with a torsional energy system and escapement mechanism, modeling components in CAD and iterating through prototypes to achieve reliable timekeeping
- **Tensile Test Machine:** Designed and built a tensile test machine using a load cell and Arduino to measure stress-strain behavior of multiple materials, identifying of material properties through data acquisition and analysis
- **Rubik's Cube Solver:** Developed a Rubik's Cube solving robot using a color sensor for initial state detection, stepper motors for precise actuation, and algorithmic solution planning to autonomously solve the cube
- **Portfolio Website:** Designed and deployed an interactive portfolio website using React and JavaScript to showcase engineering projects, technical experiences, and CAD models in a structured, user-friendly format

SKILLS

Technical: CAD, FEA, Electrical Schematics, PCB Design, GD&T, Fluid Dynamics, Control Systems

Tools: 3D Printing (FDM, Resin), Laser Cutter, CNC Mill, Microcontrollers, PCB Assembly, Transducers, Soldering

Programming: MATLAB, C/C++, React, JavaScript, HTML

Software: Fusion 360, SolidWorks, Inventor, NX, ArduPilot, ANSYS, Arduino IDE, VS Code